

EXP.NO: 4. Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.

AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. Input:

- The user fills out a web form with their name and age and submits it.

2. Form Handling and Servlet Processing:

1. Create the Web Form:

- o Create an HTML form with two input fields: one for the user's name and one for the user's age.
- o Add a submit button to the form.
- o Set the form's action attribute to the URL of the Servlet and the method to POST (or GET).

2. Create the Servlet Class:

- o Write a Servlet that processes the form data submitted by the user.
- o The Servlet will override the doPost() (or doGet()) depending on the form method) to handle form submissions.
- o The Servlet will retrieve the name and age parameters from the request object using request.getParameter("name") and request.getParameter("age").

3. Process the Data in the Servlet:

- o Extract the name and age values submitted from the form.
- o Validate the input (e.g., check if the age is a valid number).

- o Prepare a response that will display the user's name and age back on the browser.

4. Send Response to the Browser:

- o The Servlet will send a response to the browser containing a message like:

- "Hello, [name]! You are [age] years old."

- o The response will be displayed to the user in the browser.

5. Deploy the Servlet:

- o Compile the Servlet and deploy it to a Servlet container (like Apache Tomcat).

- o Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the form's action attribute.

6. Test the Application:

- o Open the web browser, fill in the form with the name and age, and submit it.

- o The Servlet will process the data and display the message back to the user in the browser.

Code:

```
package com.demo;
```

```
import jakarta.servlet.*;
```

```
import jakarta.servlet.http.*;
```

```
import jakarta.servlet.annotation.*;
```

```
import java.io.*;
```

```
@WebServlet("/UserServlet")
```

```
public class UserServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
        String age = request.getParameter("age");

        out.println("<html>");

        out.println("<body style='font-family:Arial;text-align:center;background:#f4f4f4;'>");

        out.println("<div style='margin-top:120px;background:white;padding:30px;border-
            radius:10px;width:350px;margin-left:auto;margin-right:auto;box-shadow:0 5px 15px rgba(0,0,0,0.2);'>");

        out.println("<h2>User Details</h2>");

        out.println("<p><b>Name:</b> " + username + "</p>");
        out.println("<p><b>Age:</b> " + age + "</p>");

        out.println("<a href='form.html'>Go Back</a>");
```

```
out.println("</div>");
```

```
out.println("</body>");
```

```
out.println("</html>");
```

```
}
```

```
}
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>User Information</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<div class="card">
```

```
<h2>User Information Form</h2>
```

```
<form action="UserServlet" method="post">
```

```
<label>User Name</label>
```

```
<input type="text" name="username" placeholder="Enter your name" required>
```

```
<label>Age</label>
```

```
<input type="number" name="age" placeholder="Enter your age" required>
```

```
<button type="submit">Submit</button>
```

```
</form>
```

```
</div>
```

```
</body>
```

```
</html>
```

```
body{
```

```
font-family: Arial, sans-serif;
```

```
background: linear-gradient(135deg,#4facfe,#00f2fe);
```

```
height:100vh;
```

```
display:flex;
```

```
justify-content:center;
```

```
align-items:center;
```

```
}
```

```
.card{
```

```
background:white;

padding:40px;

border-radius:10px;

width:320px;

box-shadow:0 8px 20px rgba(0,0,0,0.2);

}
```

```
h2{

text-align:center;

margin-bottom:20px;

}
```

```
label{

font-weight:bold;

}
```

```
input{

width:100%;

padding:10px;

margin:8px 0 15px;

border-radius:5px;

border:1px solid #ccc;

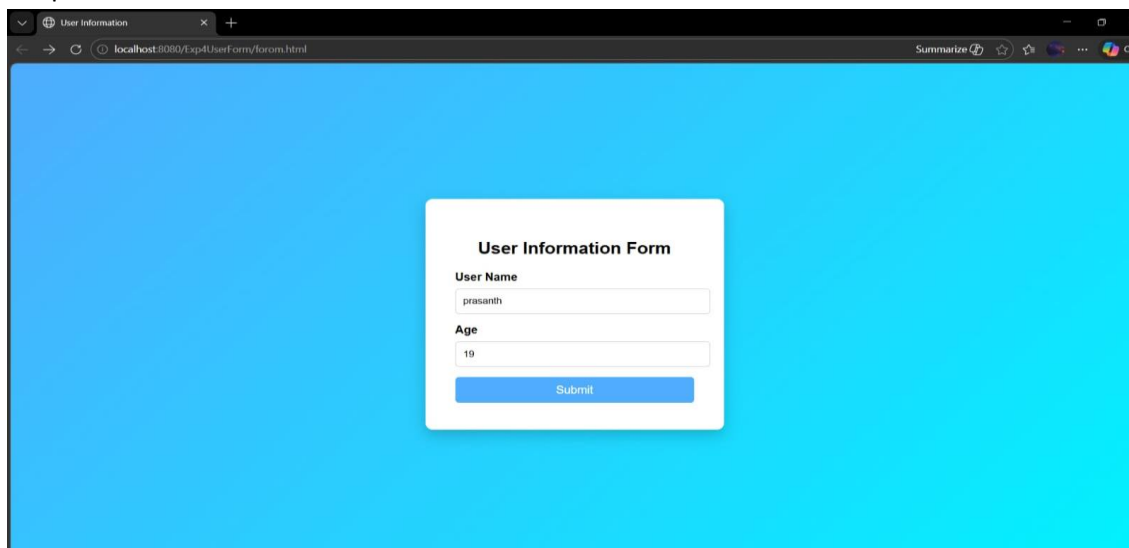
}
```

```
button{
```

```
width:100%;  
padding:10px;  
background:#4facfe;  
border:none;  
color:white;  
font-size:16px;  
border-radius:5px;  
cursor:pointer;  
}
```

```
button:hover{  
background:#0077ff;  
}
```

Output:



RESULT:

A Servlet is created to process the HTML form data and display it back on the browser.